

Email Subject Line Generation

Project Report - Group 12

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Contents

1	Introduction.	2
2	Problem Description	2
3	Dataset	2
4	Data Preprocessing	2
5	Models Used	2
6	Zero-Shot Testing	3
7	Model Fine-Tuning and Evaluation	3
8	Performance Metrics	3
9	Model Weights	5
10	Observations and Further Reading	5
11	Gradio App Integration	5
12	Challenges	5
13	Conclusion	5

1 Introduction

In today's digital communication landscape, email remains a critical tool for both personal and professional correspondence. However, crafting concise, relevant, and engaging subject lines for emails can be time-consuming, especially when handling large volumes of messages. This project aims to address this challenge by leveraging the concept of abstract summarization to automatically generate appropriate email subject lines from the body of the email content.

The project utilizes an encoder decoder model based on deep learning to perform this task, showcasing how natural language processing (NLP) can streamline communication workflows

2 Problem Description

This project proposes a solution using abstract summarization techniques, where a model learns to generate subject lines by understanding and rephrasing the core message of the email content. The goal is to improve the efficiency, consistency, and quality of email subject generation through natural language understanding and generation.

3 Dataset

We used the AESLC (Annotated Enron Subject Line Corpus) dataset, available on GitHub AESLC. The dataset includes cleaned, filtered, and deduplicated emails from the Enron Email Corpus and is split into training and test sets:

- Training set: 14,436 emails
- Test set: 1,906 emails
- Average email length: 75 words
- Average subject length: 4 words

Each email in the train and test datasets contains a subject line denoted as @subject

4 DataPreprocessing

We performed the following preprocessing steps on the data:

- Removed unnecessary spaces and replaced multiple spaces with a single space.
- Filtered out non-word and non-space characters.
- Extracted the content before @subject as the email body.
- Extracted @subject as potential subject lines.
- Combined the datasets into single files for training and testing, with the following

5 ModelsUsed

We fine-tuned and evaluated several models for the task. The following table provides an overview of the models used:

Model	Training Notebook
BART & T5Small	T5 & BART Model.ipynb
PEGASUS	Pegasus Model.ipynb

Table 1: Models used for training and testing

6 Zero-Shot Testing

We performed zero-shot testing on each of the models and observed a common issue: the generation of subject lines that were excessively long. To mitigate this, we experimented with several hyperparameters to better control the output length.

- length penalty
- max new tokens
- max length
- min length
- number of beams

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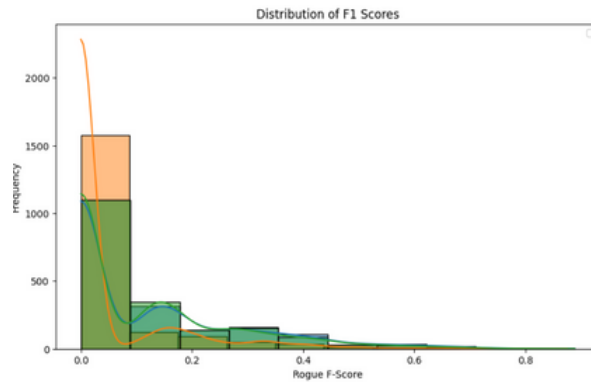


Fig 1: Histogram of Zero Shot testing of T5 small model using rouge metric

7 Model Fine-Tuning and Evaluation

We have fine tuned the three models and observed that compared to zero shot, all model are delivering email subject analogous to email context instead to simply extracting sentences. For evaluation, we used Rouge, cosine similarity using SBERT and human evaluation of random 100 sample.

8 Performance Metrics

We evaluated the performance of each model using the ROUGE score, which measures the overlap between the generated and reference subject lines and cosine similarity using SBERT. The following table provides a comparison of the ROUGE scores across different models:

Model	ROUGE 1	ROUGE 2	ROUGE L	SIMILARITY
BART	0.3313	0.1700	0.3251	0.4878
T5 SMALL	0.2865	0.1377	0.2808	0.4805
PEGAS US	0.2938	0.1433	0.2872	0.4784

Table 2: Rouge and Similarity score comparison for different models

Below is histogram plot of rouge metric, cosine similarity evaluation of test dataset and count plot of manual evaluation of random 100 samples for each model starting from T5-small, BART and Pegasus fine tuned model.

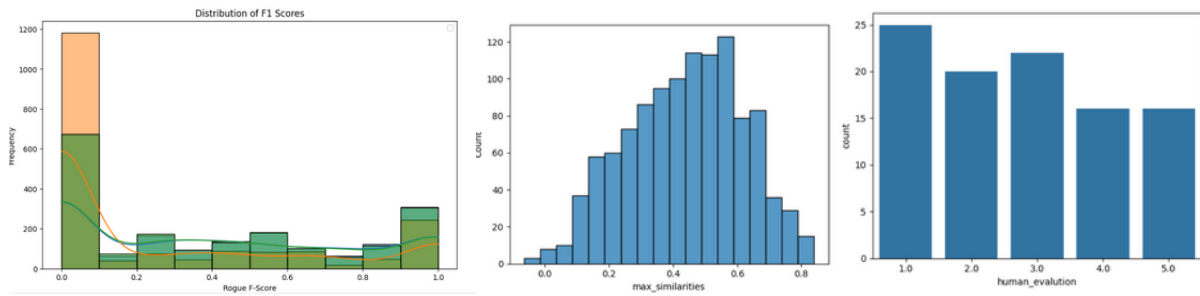


fig2: Histogram of ROUGE, cosine similarity and count plot of human evaluation of t5 small modal

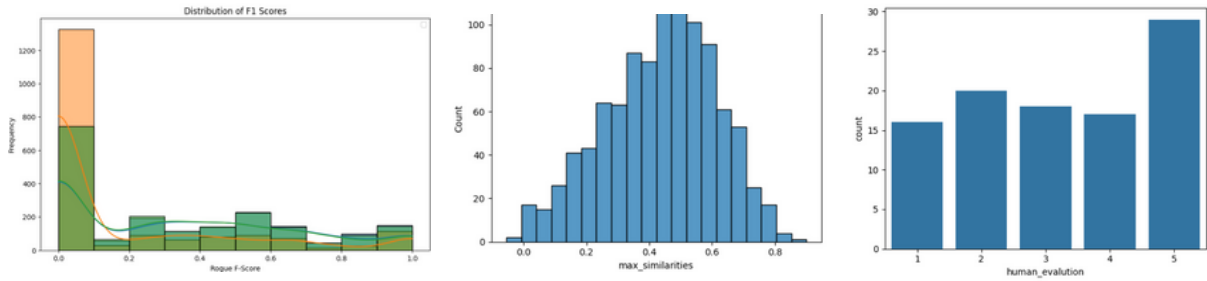


fig3: Histogram of ROUGE, cosine similarity and count plot of human evaluation of BART modal

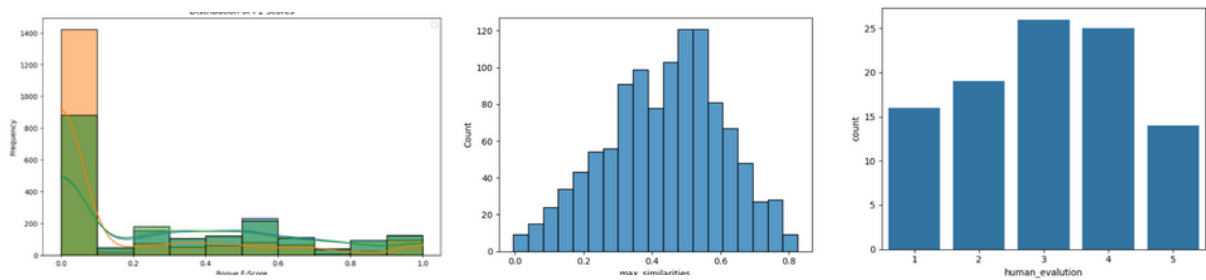


fig4: Histogram of ROUGE, cosine similarity and count plot of human evaluation of Pegasus modal

9 ModelWeights

The fine-tuned weights for each of the models were saved on GITHUB for further use:

- BART: BART on GITHUB
- T5 Small: T5 Small on GITHUB
- Pegasus : Pegasus on GITHUB

10 Observations and Further Reading

We observed that BART slightly performed better than other model in all evaluation metrics because BART is denoising autoencoder which directly improves its text regeneration ability. It is also a extractive type model which benefits ROUGE. Bart was trained on CNN/Dailymail dataset which is extractive type dataset. Model size also contribute to richer output. Moreover T5 and Pegasus are tend to abstractive which falls out when using ROUGE metric

11 Gradio App Integration

Model is deployed using Gradio

- BART: BART Gradio App
- T5: T5 Gradio App

12 Challenges

Model under performs when there is excess technical details or it has email context out of training context. Subject predication are tend to be extractive which sometimes misses to important keyword which could further improve human evaluation score. AESLC dataset has both personal and financial term which can easily fill up vocabulary. Model like Pegasus and BART is very computational and time consuming.

13 Conclusion

It was a great learning experience fine tuning these transformer models to perform a specific task. The nuances of encoder-decoder models and how it could influence the performance was a great learning experience